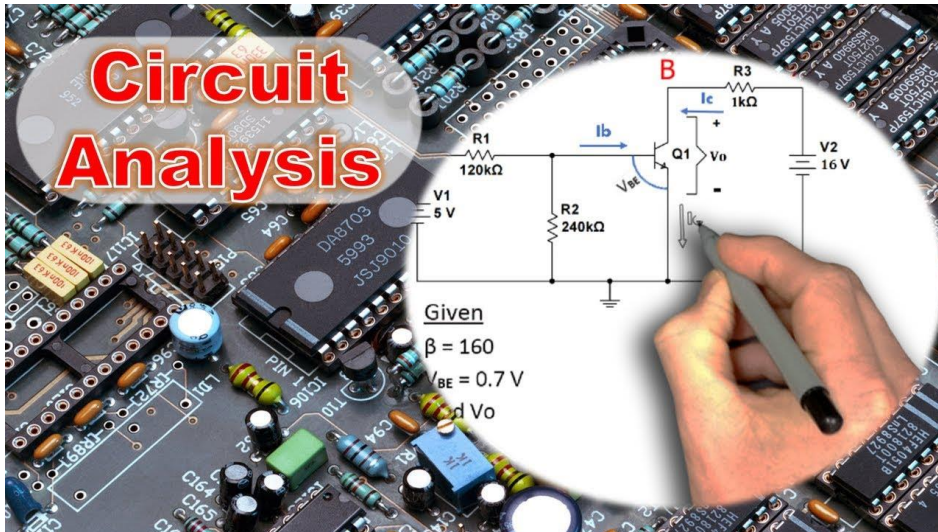




CIRCUIT ANALYSIS USING DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORMS

ABSTRACT:

This project focuses on analyzing RLC circuit, using Second Order differential equations. We begin with Homogeneous circuits examining natural responses and damping behavior .Then we extend to non –homogeneous circuit, solving them using Laplace transform to handle various input types. The input studied include , impulse, Gaussian ,and piecewise signals .we also explore RL ,RC and RLC Circuits under both DC and AC excitation ,highlighting how different configurations and inputs effect circuit behaviour.Our goal is to understand and compare how electrical circuits responds to various excitation using mathematical models .

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COURSE: Applied Mathematic –BSAM_04

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Date: 09_09_2025

Homogeneous RLC Circuit Equation:

$$LQ''(t) + RQ'(t) + \frac{1}{C}Q(t) = 0$$

This comes from Kirchhoff's voltage Law with no external source $V(t) = 0$, which means the current is free (no input voltage after initial energy is stored).

Step1: Assume the solution of the form: $Q(t) = e^{rt}$

This form makes differentiation easy and fits nicely into linear ODE, with constant coefficients.

i) $Q(t) = e^{rt}$

ii) $Q'(t) = re^{rt}$

iii) $Q''(t) = r^2e^{rt}$

Substitute i, ii, and iii into the homogeneous equation:

$$Lr^2 e^{rt} + Rre^{rt} + \frac{1}{C}e^{rt} = 0$$

Factor out $e^{rt} \neq 0$

$$e^{rt}(Lr^2 + R + \frac{1}{c} = 0$$

Step 2: Solve the characteristic equation:

$$Lr^2 + R + \frac{1}{c} = 0$$

This is a quadratic equation in r:

$$\text{Let's call it: } r^2 + \frac{R}{L}r + \frac{1}{Lc} = 0$$

Use the quadratic formula:

$$r = -\frac{\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2}$$

Let's define the discriminant:

$$\Delta = \left(\frac{R}{L}\right)^2 - \frac{4}{LC}$$

Step3: Analyze the discriminant Δ

There are three cases:

1: Overdamped: $\Delta > 0 \rightarrow$ Two real and distant roots

Solution:

$$Q(t) = Ae^{r_1 t} + Be^{r_2 t}$$

2: Critically damped: $\Delta = 0$ one repeated real root

3: Underdamped: $\Delta < 0$ Complex roots

Solution:

$$Q(t) = e^{\alpha t} (A \cos(\beta t) + B \sin(\beta t)) \text{ Where } \alpha = \operatorname{Re}(r), \beta = \operatorname{Im}(r)$$

Non –Homogeneous RLC Circuit Equation:

Now consider the non _homogeneous Equation

$$LQ''(t) + RQ'(t) + \frac{1}{C}Q(t) = V(t)$$

To solve this, we use **method of undetermined coefficients**.

The solution consist of two parts:

1: The Homogeneous Solution, $Q(t)$, which we already found.

2: The Particular Solution, $Q_p(t)$, which depend on the form of $V(t)$. The Particular Solution, $Q_p(t)$ which dep.

- If $v(t)$ is a constant (like step input) –assume $Q_p(t) = k$, where k is a constant
- If $V(t)$ is exponential (like $V(t) = Ae^{\lambda t}$, assume $Q_p(t) = Be^{\lambda t}$
- If $V(t)$ is a Sine or cosine (like $V(t) = A \cos(wt) + B \sin(wt)$, assume $Q_p(t) = C \cos(wt) + D \sin(wt)$

For such cases, substitute $Q_p(t)$ into the differential equation and Solve for the undetermined coefficients

✓ General Solution

The general solution to the non-homogeneous:

$$Q(t) = Q_h(t) + Q_p(t) \quad .$$

Where $Q_h(t)$ is the solution to the homogeneous equation and $Q_p(t)$ is the particular Solution.

- **Apply the initial condition :**

To fully determine the constants $C1$ and $C2$ we use the initial conditions: This might be the initial charge and initial current, which are typically given as:

$$Q(0) = Q_o, \quad Q'(0) = 1$$

Commented [CM1]:

We can use the initial conditions to solve for the constants C1 and C2 in the general Solution.

This approach will help us to provide the complete Solution for the Second _ Order non-homogeneous differential equation in the context of circuit Analysis.

- **Here we use Laplace Transform to Solve the second ODE**

To solve second _order differential equation (2ODE) in a Dc circuit, using Laplace Transform, Lets walk through a common example: Series RLC circuit with a Dc voltage Source.

Circuit Description:

A series RLC circuit with a Dc Voltage Source V_o , connected at $t = 0$.

V_o (DC) $\rightarrow$ [R] $\rightarrow$ [L] $\rightarrow$ [C] $\rightarrow$ Ground.

Let:

R= Resistance

L= inductance (H)

C = Capacitance (p)

q(t) = charge on capacitor at time t

$i(t) = \frac{dq}{dt}$ = current

Write the 2ODE from KVL

Using Kirchhoff's Voltage Law:

$$V_0 = Ri(t) + L \frac{di}{dt} + \frac{1}{C} q(t)$$

Substitute: $i(t) = \frac{dq}{dt}$

$$V_0 = R \frac{dq}{dt} + L \frac{d^2q}{dt^2} + \frac{1}{C} q(t)$$

Rewriting:

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{c} q(t) = V_0$$

- Take Laplace Transform of Both Sides.

Let $Q(s) = \text{Laplace } \{q(t)\}$. Assume two initial conditions: $q(0), q'(0) = 0$

Apply Laplace transforms:

- $\mathcal{L} \left\{ \frac{d^2 q}{dt^2} \right\} = s^2 Q(s)$
- $\mathcal{L} \left\{ \frac{dq}{dt} \right\} = sQ(s)$
- $\mathcal{L}\{q(t)\} = Q(s)$

Apply to the ODE:

$$Ls^2 Q(s) + RsQ(s) + \frac{1}{c} Q(s) = \frac{V_0}{s}$$

Factor out $Q(s)$:

$$Q(s) = \left(Ls^2 + Rs + \frac{1}{C} \right) = \frac{V_o}{s}$$

✓ Solve for Q(s)

$$Q(s) = \frac{V_o}{s \left(Ls^2 + Rs + \frac{1}{C} \right)}$$

✓ Take inverse Laplace to find q(t) or i(t)

We can :

- Do partial Decomposition on Q(s) , then invert to get q(t)
- Or differentiate q(t) to get current

$$I(t) = \frac{dq}{dt}$$

Let pick values so we can proceed with the inverse Laplace Transform.

Assume:

- $V_o = 12V$
- $= 12\text{ v}$
- $R = 4\text{ ohms}$
- $L = 1H$
- $C = 1/2\text{ F}$

Then:

$$Q(s) = \frac{12}{s(s^2 + 4s + 2)}, \text{ we need to find } I(t) = dq/dt$$

Find the inverse Laplace of $Q(s)$

$$\rightarrow Q(s) = \frac{12}{s(s^2 + 4s + 2)}$$

$$\rightarrow \frac{12}{s(s^2 + 4s + 2)} = \frac{A}{s} + \frac{Bs + c}{s^2 + 4s + 2}$$

Multiply both sides by:

$$12 = A(s^2 + 4s + 2) + (Bs + C)s$$

Expand right hand side:

- s^2 terms : $A+B = 0$
- s terms : $4A + c = 0$
- Constant : $2A = 12 \dots A = 6$

Now solve $A = 6$

$$A+B = 0 \dots B = -6$$

$$4A + C = 0 \dots c = -24$$

So:

$$Q(s) = \frac{6}{s} + \frac{-6s - 24}{s^2 + 4s + 2}$$

➤ **Modeling: Electric RLC Circuit**

Designing a good model is a task the computer cannot do.

Designing good models is a task the computer cannot do. Hence setting up models has become an important task in modern applied mathematics. The best way to gain experience in successful modeling is to carefully examine the modeling process in various fields and applications. Figure 61 shows an **RLC**-circuit, as it occurs as a basic building block of large electric networks in computers and elsewhere. An RLC-circuit is obtained from an RL-circuit by adding a capacitor. The model of the RL-circuit is $LI' + RI = E(t)$. It was obtained by **KVL** (Kirchhoff's Voltage Law) by equating the voltage drops across the resistor and the inductor to the EMF (electromotive force). Hence we obtain the model of the RLC-circuit simply by adding the voltage drop across the capacitor. Here, C F (farads) is the capacitance of the capacitor. Q coulombs is the charge on the capacitor, related to the current by

$I(t) = dQ/dt$, equivalently $Q(t) = \text{integral } I(t)$

Name	Symbol	Notation	Unit	Voltage Drop
Ohms Resistor		R ohms Resistance	ohms()	RI
Inductor		L Inductance	henrys(H)	$L(di/dt)$
Capacitor		C Capacitance	Farads (F)	Q/C

This text walks through the process of solving the non-homogeneous Second _ order differential equation that governs the current $I(t)$ in an RLC Circuit with a sinusoidal electromotive force .

Governing Equation:

$$I(t) = \frac{dQ}{dt}, \epsilon(t) = \epsilon_o \sin (vt)$$

From Kirchhoff's voltage Law

$$L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = \epsilon(t)$$

Since $I(t) = \frac{dQ}{dt}$, differentiate both sides gives :

$$L \frac{d^2I}{dt^2} + R \frac{dI}{dt} + \frac{1}{C} I = \frac{d}{dt} \epsilon(t) = \epsilon_o v \cos (vt)$$

Step: Non homogeneous ODE

The equation becomes:

$$LI'' + RI' + \frac{1}{C}I = \epsilon_0 v \cos(vt) \dots (1)$$

Find particular Solution $I_p(t)$

Use the method of undetermined coefficients .Assume:

$$I_p(t) = a \cos(vt) + b \sin(vt) \dots (2)$$

Then compute derivatives:

$$I_p' = -a v \sin(vt) + b v \cos(vt)$$

$$I_p'' = -a v^2 \cos(vt) - b v^2 \sin(vt)$$

Substitute into the equation (1):

$$L(-a v^2 \cos(vt) - b v^2 \sin(vt)) + R(-a v \sin(vt) + b v \cos(vt)) + \frac{1}{C}(a \cos(vt) + b \sin(vt)) = \epsilon_0 v \cos(vt)$$

Group Cosine and sine terms:

Cosine terms:

$$(-aL V^2 + b R v + \frac{a}{C}) \cos(vt)$$

Sine terms:

$$(-bL V^2 - aR v + \frac{b}{C}) \sin(vt)$$

Set up system.

$$(-aL V^2 + b R v + \frac{a}{C}) = \epsilon_o v \quad (\text{cosine})$$

$$(-bL V^2 - aR v + \frac{b}{C}) = 0 \quad (\text{sine})$$

Step 3: Simplifying using Reactance S

Define reactance:

$$S = vL - 1/vC$$

Now rewrite the system:

$$R_a + S b = 0 \dots\dots\dots (3)$$

$$Sa - Rb = \epsilon_o \dots\dots\dots (4)$$

Multiplying (3) by S, (4) by R, then add;

$$RSa + S^2 b + S^2 b + S^2 a - R^2 b = S\epsilon_o \dots\dots (R^2 + S^2) a = -\epsilon_o S \dots\dots a = (-\epsilon_o S / (R^2 + S^2))$$

Now eliminate a: multiply (3) by R, (4) by S, then subtract

$$R^2 a + RSb, S^2 a - RSb = R\epsilon_o \dots\dots (R^2 + S^2) b = \epsilon_o R \dots\dots b = \frac{\epsilon_o R}{R^2 + S^2}$$

Particular Solution

Plug into (2) :

$$I_p(t) = a \cos(vt) + b \sin(vt) = \frac{-\epsilon_o S}{R^2 + S^2} \cos(vt) + \frac{\epsilon_o R}{R^2 + S^2} \sin(vt)$$

Express in Sinusoidal Form:

Use identity:

$$A \cos(vt) + B \sin(vt) = I_o \sin(vt - \phi)$$

Where:

$$I_o = \frac{\epsilon_o}{\sqrt{R^2 + s^2}} \text{ (amplitude)}$$

$$\tan(\phi) = S/F \text{ (phase lag)}$$

$$\text{So: } I_p(t) = I_o \sin(vt - \phi) \dots\dots\dots(5)$$

Step 4: General Solution:

Homogeneous equation

$$LI'' + RI' + \frac{1}{C}I = 0$$

Characteristic equation:

$$Lr^2 + Rr + \frac{1}{C} = 0 \dots\dots r = \frac{-R \pm \sqrt{R^2 - 4L/C}}{2L}$$

General homogeneous solution:

$$I_h(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

Since $R > 0$, both roots are negative real or complex with negative real parts $I_h(t) \rightarrow 0$ as $t \rightarrow \infty$

Final General Solution >

$$I(t) = I_h(t) + I_p(t) = \text{transient} + \text{steady state} \dots\dots I(t) \dots I_o \sin(\omega t - \phi)$$

- ✓ The RLC circuit leads to a second _ order non-homogeneous ODE
- ✓ The particular solution represent the steady state current due to the sinusoidal input
- ✓ The homogeneous solution represent transient behavior that fades with time

- ✓ Final behavior is a sinusoidal current with same frequency as the input but phase shifted and possibly with reduced amplitude due to resistance and reactors

Modeling RL _ Circuit

Model the RL _ Circuit shown in the figure below and solve the resulting ODE for the current $I(t)$. A (amperes) , where t is the time . Assume that the circuit contain as an EMF $E(t)$ (Electromotive force) a battery of $E = 48$ volts , which is constant , a Resistor of $R = 11$ ohms , and an inductor of $L = 0.1$ H (henrys) , and that the current is initially zero (o) .

Physical Laws : A current in the circuit causes a voltage drop RI across the resistor (ohms law) and voltage drop $LI' = L(di/dt)$ across the conductor and sum of these two voltage drop equals the EMF (Kirchhoff's voltage Law , KVL) .

- ❖ Remarks: In general, kvl states that “The voltage (Electromotive force EMF) impressed on a closed loop is equal to the sum of voltage drop across all the other elements of the loop “. For Kirchhoff's current Law (kcl) and historical information.

Solution: According to these laws the model of the RL_Circuit is $LI' + RI = E(t)$, in standard form.

$$I' + R/L I = E(t) / L$$

We can solve this linear ODE by using the formula;

$$Y(x) = e^{-h} \left(\int e^h r dx + c \right) \quad h = \int p(x) dx$$

We let $x=t$, $y=I$, $p=R/L$, $h=(R/L)t$, obtain the general

Solution :

$$I = e^{-\left(\frac{R}{L}\right)t} \left(\int e^{\left(\frac{R}{L}\right)t} \left(\frac{E(t)}{L} \right) dt + c \right)$$

By integration:

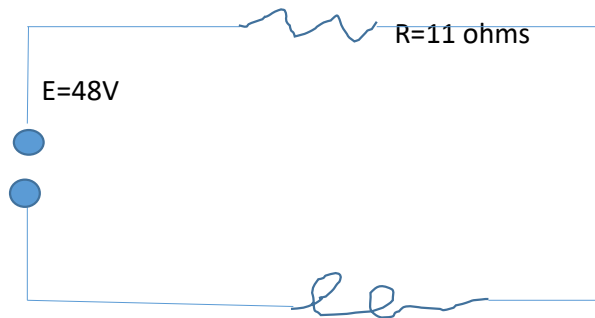
$$I = e^{-\left(\frac{R}{L}\right)t} \left(\frac{\frac{E}{L} e^{\left(\frac{R}{L}\right)t}}{\frac{R}{L}} + c \right) = \frac{E}{R} + c e^{-\left(\frac{R}{L}\right)t}$$

For our case, $R/L = 11/0.1 = 110$ and $E(t) = 48/0.1 = 480 \neq 0$ const ;

Thus,

$$I = \frac{48}{11} + ce^{-110t}$$

The figure of RL _ Circuit.



$$L=0.1H$$

- What is step input?

A step input is a sudden application of voltage that remains constant overtime .It is mathematically modelled as:

$$V(t) = V_o * U(t)$$

Where:

- V_o is the step voltage (a constant)
- $U(t)$ is the Heaviside step function

When is it used?

- ✓ To stimulate Turning on a Dc power supply instantly
- ✓ In analyzing startup behavior of a circuits and systems.
- ✓ Common in testing transient response of fitters and amplifiers

- **General Second Order ODE (Series RLC _ Circuit)**

Using Kirchhoff's law Voltage law (kvl) , we get :

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = V_{oo}$$

This is a non-homogeneous second order ODE with constant forcing term V_{oo} . We solve it in two parts:

1: Homogeneous Solution (Transient Response):

$$L \frac{d^2 Q_h}{dt^2} + R \frac{dQ_h}{dt} + \frac{1}{C} Q_h = 0$$

Solve using characteristic equation:

$$Lr^2 + Rr + \frac{1}{C} = 0$$

Roots, r_1, r_2 , define the transient behavior:

- Overdamped

- Critically damped
- Under damped

2: Particular Solution (Steady state Response)

Since $V(t) = V_o$ is constant, assume a constant solution

$$Q_p(t) = Q_\infty = CV_o$$

This give a steady state Voltage across the capacitor and steady current of zero (after infinite time) .

General solution:

$$Q(t) = Q(t) + Q_p(t).$$

- What is an impulse input?

An impulse input is an idealized, very short burst of voltage (or current) applied to a circuit. It modelled by the Dirac delta function:

$$V(t) = \delta(t)$$

- ✓ Physically: It represent an instantaneous “kick” to the system at $t=0$ with a very high amplitude but infinitesimally short duration.
- ✓ Mathematically : It hat the property

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

When is it used?

Impulse input are widely used to:

- Analyze the system response _ especially impulse response .which helps define how the system behave overtime.
- Test system for transient behavior
- In Laplace transform , it is useful due to its transforming being 1

ODE for impulse input:

Start with the standard _ Order ODE for a series RLC circuit:

$$L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = V(t)$$

For an impulse input:

$$V(t) = \delta(t)$$

So the equation becomes:

$$L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = \delta(t)$$

Take the Laplace transform.

Assume initial conditions:

- $Q(0^+) = 0$
- $\frac{dQ}{dt}(0^+) = 0$

Take Laplace transform on both sides:

- $\text{Laplace}\{d^2Q/dt^2\} = s^2 Q(s)$
- $\text{Laplace}\{dQ/dt\} = sQ(s)$
- $\text{Laplace}\{\delta(t)\} = 1$

So the transformed equation becomes:

$$LS^2Q(s) + RsQ(s) + \frac{1}{C} Q(s) = 1$$
$$\rightarrow Q(s) \left(LS^2 + Rs + \frac{1}{C} \right) = 1$$

$$\rightarrow Q(s) = \frac{1}{Ls^2Rs + \frac{1}{c}}$$

Inverse Laplace transform

Now inverse Laplace transform $Q(s)$. First define:

$$\rightarrow Q(s) = \left(\frac{1}{L \left(s^2 + \frac{Rs}{L} + \frac{1}{LC} \right)} \right)$$

Let:

- $\alpha = \frac{R}{2L}$
- $\omega_0^2 = \frac{1}{LC}$

- Then the denominator becomes :
- $s^2 + 2\alpha s + \omega_0^2$

We now consider three cases:

(a) : Overdamped $\alpha > \omega_0^2$

Inverse Laplace gives:

$$Q(t) = \frac{1}{L} t e^{-\alpha t} \propto t$$

(b) : Critically Damped : $\alpha = \omega_0^2$

Inverse Laplace gives:

$$Q(t) = \frac{1}{L} t e^{-\alpha t} \propto t$$

(c): Underdamped $\alpha < \omega_0^2$

Let

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Then inverse Laplace gives:

Q (t) =

$$\frac{1}{L\omega_d} e^{-\alpha t} \sin(\omega_d t)$$

This is decay oscillation –the transient response from the impulse.

✓ **Get current I (t)**

Since $I(t) = \frac{dQ}{dt}$, differentiate the result accordingly.

For example, underdamped:

Q (t) =

$$\frac{1}{L\omega_d} e^{-\alpha t} \sin(\omega_d t)$$

Then:

$I(t) =$

$$\frac{dQ}{dt} = \frac{1}{L} e^{-\alpha t} \left[\cos(\omega_0 t) - \frac{\alpha}{\omega_d} \sin(\omega_0 t) \right]$$

The current response $I(t)$ of the RLC Circuit to an impulse input is:

- A pure transient response _ no steady state
- Varies base on the damping condition :
 - I. Over damped : two exponential
 - II. Critically damped : exponential * Time
 - III. Underdamped : decaying oscillations

What is a Gaussian input?

The Gaussian function is given by:

$$V(t) = Ae^{\frac{-(t-t_o)^2}{2Q^2}}$$

- **A** : Peak Amplitude
- **t_o** : Center of the pulse (time of which peak occurs)
- **Q** : Width of the pulse (control how narrow or wide it is) where it's used :
 - Signal processing : Modelling localized signals like pulse in radar and communication systems
 - Control systems : Approximate impulse response (smooth version of delta function)
 - Physics and electronics : Optical and RF system often use Gaussian signals to model light waves
 - The governing ODE (series RLC example)

We assume a series RLC circuit. The ODE becomes:

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = V(t)$$

Substitute the Gaussian input:

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = A e^{\frac{-(t-t_0)^2}{2\sigma^2}}$$

Solve the homogeneous equation:

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = 0$$

Use the characteristic equation:

$$Lr^2 + Rr + \frac{1}{C} = 0$$

Solve for roots → find homogeneous solution $Q_h(t)$

2: Solve the particular solution:

Because the right hand side is a Gaussian, undetermined coefficient won't work well. Instead we use method of variation of parameters or convolution with the impulse response.

➤ What is a piecewise input?

..... A piecewise input means the voltage source behave differentially in different time intervals:

A simple example:

$$V(t) = \begin{cases} A, & 0 \leq t < t_1 \\ B, & t \geq t_1 \end{cases}$$

This kind of input is useful for modelling switching behavior, such as:

- Changing power supply level
- On _off control System
- Step changes in applied Voltage

When is piecewise input used?

Piecewise signals are commonly used in:

- Switching circuits
- Digital logic _ driven analog systems.
- Pulse _ width modulation
- Control system with logic states (E.g. Heater ON for $t < t_1$,Off for $t > t_1$)

These signals let us steady how the circuit responds to sudden transitions between different constant or varying levels.

Solving the ODE for a Piecewise Input

....Lets use the series RLC circuit Again:

Step 1: Governing Equation

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q(t) = V(t)$$

Now let:

$$V(t) = V(t) = \begin{cases} A, & 0 \leq t < t_1 \\ B, & t \geq t_1 \end{cases}$$

We solve this equation into two parts

- Solve for $0 \leq t < t_1$,
- The equation becomes :

$$L \frac{d^2}{dt^2} + R \frac{dq_1}{dt} + \frac{1}{C} q_1 = A$$

Assume:

$$q_{p_1}(t) = Q_1 \quad \dots \text{Constant solution}$$

Substitute:

$$0 + 0 + \frac{1}{C} Q_1 = A \quad \dots Q_1 = AC$$

So the full solution is:

$$q_{1(t)=Q_h}(t) + AC$$

Use initial condition at $t=0$ to find constant in $q_h(t)$.

Solve for $t \geq t_1$

$$L \frac{d^2 q^2}{dt^2} + R \frac{dq^2}{dt} + \frac{1}{C} q^2(t) = B$$

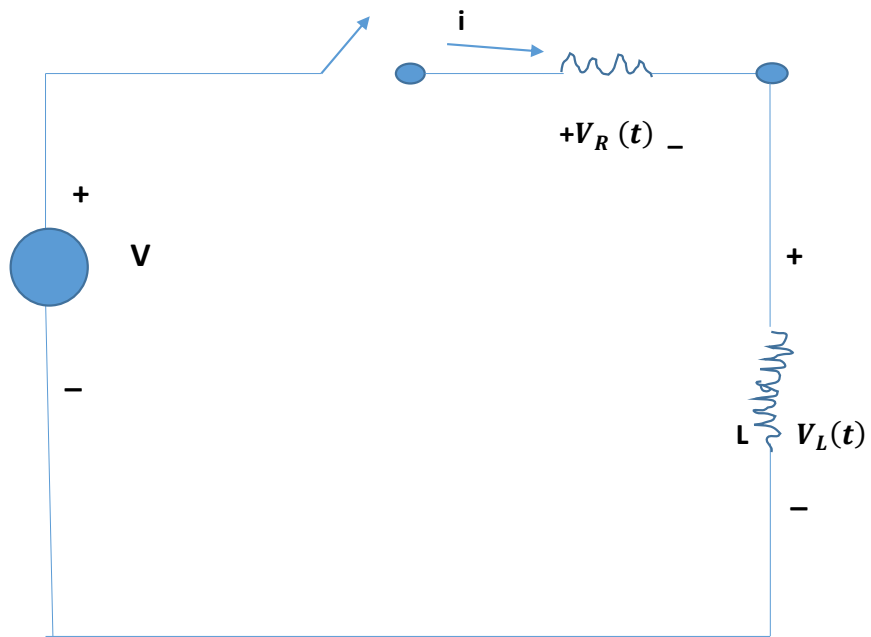
Assume:

$$q_{p_2(t)=Q_2} \quad \dots \frac{1}{C} = Q_2 = B \quad \dots = Q_2 = BC$$

So the full solution is

$$q^2(t) = q_{h_2(t)} + BC$$

RL Circuit with DC Excitation.



- initially , $i(0) = 0$ (current is zero before $t=0$)
- The inductor oppose the change in current by generating a voltage $V_L(t)$
- Current builds up over time, and voltage across the inductor decrease at the current increase.
Apply Kirchhoff's voltage Law

$$V = V_R(t) + V_L(t)$$

Using ohms law for the resistor and the inductor relationship:

$$V = Ri(t) + L \frac{di(t)}{dt} \quad \text{for } t > 0$$

- Rearrange and separate the variables :

$$L \frac{di}{dt} = V - Ri$$

$$\frac{L}{V - Ri} di = dt$$

- Integrating both sides.

$$\frac{L}{R} \ln(V - Ri) = t + k$$

Use initial condition $I(0) = 0$ to solve for k .

$$K = -\frac{L}{R} \ln(v)$$

$$\ln\left(\frac{V-Ri}{V}\right) = -\frac{Rt}{L}$$

$$\frac{V-Ri}{V} = e^{-Rt/L}$$

Thus the current $i(t)$ is:

$$i(t) = \frac{V}{R} [1 - e^{-Rt/L}] \text{ for } t > 0$$

This is the current response of the RL circuit

Time Constant T :

$$T = L/R$$

This is the time constant, which determine how quickly the current reaches its final value.

Voltage across components:

- Resistor :

$$V_r(t) = i(t) \cdot R = V[1 - e^{-t/\tau}]$$

- Inductor :

$$V_L(t) = i(t) \cdot R = V[1 - e^{-t/T}]$$

- Summary of results :
- Current $i(t) = \frac{V}{R} [1 - e^{-t/T}]$
- Resistor Voltage ($V_R - r(t)$):
 $v_R(t) = V[1 - e^{-t/T}]$
- Inductor Voltage ($V_L(t)$):
 $v_L(t) = V - v_R(t)$
- The current approaches $\frac{V}{R}$ as time goes to infinity, and resistances decreases.

RL circuit with AC excitation

- Circuit setup:
- ✓ An AC voltage source $V(t) = V_0 \sin(\omega t)$ is connects in series with a Resistor R and an inductor L
- ✓ The goal is to determine the current $i(t)$ in the circuit .
 Apply Kirchhoff's voltage Law (kvl)

$$V(t) = V_R(t) + V_L(t)$$

Where:

- $V_R(t) = Ri(t)$ voltage across the resistor
- $V_L(t) = L \frac{di(t)}{dt}$ voltage across the inductor

Substitute the expression for $V_R(t)$ and $V_L(t)$:

$$V_0 \sin(\omega t) = Ri(t) + L \frac{di(t)}{dt}$$

✓ Solve using phasor Representation

Represent the AC voltages and current using phasors

- The voltage $V(t) = V_0 \sin(\omega t)$ becomes $\tilde{V} = V_0 \angle 0^\circ$
- The current $i(t)$ will have the form $\tilde{I} = I \angle \phi$

The phasor form of the equation is:

$$\tilde{V} = \tilde{I} (R + j\omega L)$$

Where j is the imaginary unit, and the impedance of the RL circuit is:

$$Z = R + j\omega L$$

Solve for the phase current $\tilde{I}$:

$$\dots \tilde{I} = \tilde{V} / Z = \frac{V_0 \angle 0^\circ}{(R + j\omega L)}$$

$$\dots \tilde{I} = \frac{V_0}{R + j\omega L} \angle -\theta_z$$

In polar form $z = \sqrt{R^2 + (\omega L)^2} \angle \theta_z$

Where phase angle θ_z is

$$\theta_z = \tan^{-1} \left(\frac{\omega L}{R} \right)$$

Convert back to time Domain:

The time domain current is :

$$i(t) = I_0 \sin(\omega t - \theta_z)$$

$$\text{Where: } I_0 = \frac{V_0}{\sqrt{R^2 + \omega^2 L^2}}$$

And the phase angle θ_z is :

$$\theta_z = \tan^{-1} \left(\frac{\omega L}{R} \right)$$

- Summary of results

➤ Current $i(t)$:

$$i(t) = \frac{V_0}{\sqrt{R^2 + \omega^2 L^2}} \sin(\omega t - \tan^{-1} \left(\frac{\omega L}{R} \right))$$

➤ Impedance of the RL circuit :

$$Z = \sqrt{R^2 + \omega L^2} \quad \angle \tan^{-1} \left(\frac{\omega L}{R} \right)$$

➤ Current Amplitude I_0 :

$$I_0 = \frac{V_0}{\sqrt{R^2 + \omega L^2}}$$

RC Circuit with External DC Excitation

Circuit Setup:

- ✓ A DC Voltage Source V is applied to a series circuit consisting of a resistor R and capacitor C .
- ✓ The capacitor is initially uncharged i.e. $V_C(0) = 0$
- ✓ When the switch is closed at $t=0$, the capacitor charges and both transient voltage and current are observed.

Apply Kirchhoff's Voltage Law

The sum of voltage around the loop gives:

$$V = V_R(t) + V_C(t)$$

Where:

- $V_R(t) = i(t) R$ (voltage across the resistor).
- $V_C(t) = \frac{1}{C} \int i(t) dt$ (voltage across the capacitor)

Using $i(t) = C \frac{dv_c(t)}{dt}$, Substitute into the kvl equation

$$\dots V = RC \frac{dv_c(t)}{dt} + V_c(t)$$

$$\dots \frac{dv_c(t)}{v - V_c(t)} = \frac{dt}{RC}$$

Integrate both sides:

$$-RC \ln(V - V_c(t)) = t + K$$

Use initial condition $V_c(0) = 0$ to find K:

$$K = -RC \ln(V)$$

Substitute k into the equation:

$$-RC \ln(V - V_c(t)) = t - RC \ln(V)$$

Simplifying:

$$\ln\left(\frac{V - V_c(t)}{V}\right) = -t/RC$$

Taking the antilogarithm:

$$\frac{V - v_c(t)}{V} = e^{-\frac{t}{RC}}$$

Thus the voltage across the capacitor is:

$$V_c(t) = V (1 - e^{-\frac{t}{RC}})$$

Voltage and current in the circuit.

- Voltage across the Resistor

$$V_r(t) = V - V_c(t) = V e^{-\frac{t}{RC}}$$

- Current through the circuit :

$$I(t) = C \frac{dv_c(t)}{dt} = \frac{V}{R} e^{-\frac{t}{RC}}$$

Alternatively

$$I(t) = \frac{V_r(t)}{R} = \frac{V}{R} e^{-\frac{t}{RC}}$$

- Time Constant.

The time constant $T=RC$, and it represents the time it takes for the voltage across the capacitor to reach approximately 63% of its final values.

RLC Circuit with External AC Excitation.

- The RLC circuit contain both an inductor and a capacitor along with a resistors. Unlike simple circuits with only one energy storage element (capacitor or inductor),RLC Circuits leads to Second –order differential equations when analyzing ,voltages or currents .
- These circuit require two initial condition to solve, making the analysis more complex than first order circuits.

Types of RLC Circuits

-Parallel RLC Circuits

-Series RLC circuits

Parallel RLC circuit with DC Excitation:

(1). Assumptions:

- ✓ The initial current in the inductor and voltage across the capacitor are zero
- ✓ The circuit has a resistor, inductor L , and capacitor C , with a DC voltage source V

Kirchhoff's current law (KCL)

➤ Apply KCL at the common node in the parallel RLC circuit

$$I = i_C + i_L$$

Where:

- I is the total current supplied by the voltage source v
- i_C is the current through the capacitor
- i_L is the current through the inductor

*Equation Setup

The voltage across the Capacitor $V_C(t) = V_L(t) = V(t)$. The equation relating these components is :

$$\frac{V - v(t)}{R} = \frac{1}{L} \int_0^t v(t') dt' + C \frac{dv(t)}{dt}$$

Simplifying further:

$$\frac{V}{R} = v(t) \left(\frac{1}{R} + \frac{1}{L} \int_0^t v(t') dt' + C \frac{dv(t)}{dt} \right)$$

- Differentiating both sides with respect to time leads to the second order linear differential equation.

$$C \frac{d^2 v(t)}{dt^2} + \frac{1}{R} \frac{dv(t)}{dt} + \frac{1}{L} v(t) = 0$$

- This simplifies to

$$\frac{d^2 v(t)}{dt^2} + \frac{1}{RC} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

- ✓ Characteristic Equation :

The equation is quadratic equation in terms of the operation s, where $s = d/dt$

$$\dots [s^2 + \frac{1}{RC} s + \frac{1}{LC}] = 0$$

$$\text{Therefore, } S1 = \frac{-1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

➤ General solution.

- The general form of the solution for v(t) is

$$V(t) = A1e^{s1t} + A2e^{s2t}$$

- Where A1 and A2 are the constant determine by the initial conditions.

RC Circuit with External AC Excitation Circuit Setup:

- ✓ A DC voltage source V is replaced with an AC voltage source $v(t) = V_0(\omega t)$, where :
- ✓ V_0 is the peak voltage

- ✓ ω is the angular frequency
- ✓ The circuit consists of resistor R and a Capacitor c in series.

Apply Kirchhoff's Voltage Law (KVL)

The sum of voltage around the loop gives:

$$V(t) = V_R(t) + V_C(t)$$

Where:

$$V_R(t) = i(t) R \quad (\text{voltage across the resistor})$$

$$V_C(t) = \frac{1}{C} \int i(t) dt \quad \text{voltage across the capacitor}$$

Using $i(t) = C \frac{dV_C(t)}{dt}$, substitute into the kvl equation:

$$V_0 \sin(\omega t) = i(t) R + V_C(t)$$

Using $V_C(t) = \frac{1}{C} \int i(t) dt$, rewrite the equation

$$V_0 \sin(\omega t) = RC \frac{dV_C(t)}{dt} + V_C(t)$$

Or:

$$\frac{dv_c(t)}{dt} + \frac{1}{RC} V_c(t) = \frac{V_o}{RC} \sin(\omega t) \quad .$$

Solution using the Method of Forced Response:

The Equation is a first order linear equation to solve it, assume a particular solution of the form

$$V_c(t) = A \sin(\omega t) + B \cos(\omega t)$$

Where A and B are constants to be determined

- Solve the differential Equation :
- The homogeneous solution to this equation is :

$$V_c(t) = C_1 e^{-t/RC} + C_2$$

- Particular solution (after solving methods like undetermined coefficients is :

$$V_c(t) = \frac{V_0}{\sqrt{R^2 + (\omega C)^2}} \sin(\omega t - \phi)$$

- where : $\phi = \tan^{-1}(\frac{\omega C}{R})$ is the phase angle
- The amplitude of the voltage across the capacitor is $\frac{V_0}{\sqrt{R^2 + (\omega C)^2}}$, which depend on the resistance R and the capacitance C

- Voltage across the Resistor and current.

- The voltage across the resistor is ::

$$V_R(t) = i(t)R = V_0 (\cos(\omega t) - \phi)$$

- The current in the circuit is :

$$I(t) = C \frac{dv_c(t)}{dt} = \frac{V_0 \omega C}{\sqrt{R^2 + (\omega C)^2}} (\cos(\omega t) - \phi)$$

Impedence of RLC Circuit,

The total impedance z of the circuit is given by

$$Z = \sqrt{R^2 + (\omega C)^2}$$

This leads to the following expression for voltage and current.

- Voltage across the resistor

$$V_R(t) = I(t)R$$

- Current in the circuit

$$I(t) = \frac{V_0}{Z} \sin(\omega t - \phi)$$

Where:

- Z is the impedance
- ϕ is the phase shift between the current and the Voltage .

AC RLC Circuits with AC Excitation:

- AC voltage source : $V(t) = V_o \sin(\omega t)$
- Impedance : Total impedance of a series RLC circuit is :
 $Z = R + j(\omega L - 1/\omega C)$
- Voltage across components:
 - ✓ Resistor: $V_R(t) = I(t) R$
 - ✓ Inductor : $V_L(t) = I(t) \omega L$
 - ✓ Capacitor : $V_C(t) = I(t) 1/\omega C$

Differentiate Equation for series RLC Circuit.

- The general differential equation for a series RLC circuit with AC excitation :

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = v_o \sin(\omega t)$$

This equation governs the current $i(t)$ in the circuit .

Solution of Differential Equation:

- The solution is the sum of two parts
- Natural response (Homogeneous solution): Response with the AC source, decay overtime.
- Forced Response (Particular solution) : steady state solution with Ac excitation

The total current $i(t)$ is :

$$i(t) = i_{natural}(t) + i_{force}(t)$$

Where $i_{force}(t)$ follows the form of the AC source

Impendence and Phase Angle:

- Impendence Magnitude: $I_o = \frac{V_o}{|Z|}$ where $|Z|$ is the magnitude of the impendence.

- Phase Angle

$$\tan \phi = \frac{X_L - X_C}{R}$$

The phase angle ϕ indicate whether the current leads or lags the voltage

Resonance:

Resonance condition occurs when $X_L = X_C$.i.e. $\omega_o = \frac{1}{\sqrt{LC}}$

At resonance, the impedance is purely resistive,

And the current is in the phase with the voltage

Key working steps:

Series RLC circuit (AC):

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = v_o \sin(\omega t)$$

Solution form:

The current response is given by:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Where s_1 and s_2 are the roots of the characteristic equation:

$$s^2 + \left(\frac{1}{RC}\right)s + \left(\frac{1}{LC}\right) = 0$$

Roots s_1 and s_2 are:

$$s_1 = \left(-\frac{1}{2RC}\right) + \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \quad ; \quad s_2 = \left(-\frac{1}{2RC}\right) - \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

Phase Angle Calculation:

- The phase Angle is determine by :

$$\mathbf{Tan (\theta) = \frac{X_L - X_C}{R}}$$

This indicate whether the current leads or lags the Voltage in the circuit.

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